Project 3 | TinkerCad | 2 class periods

Rocket Launch: Design Mission 🚀

Measure a real rocket launcher, design a rocket that fits it, 3D print it, and launch it!

Your mission

Our stomp launcher shoots a puff of air up a **launch tube**. A rocket slides down over the tube, and when you stomp, the air blasts it into the sky. You'll design your own rocket in TinkerCad that fits **our** launcher. Nobody will tell you the sizes. **You'll measure them yourself**, like a real engineer!

OUR LAUNCHER

We use a **stomp rocket launcher**: you stomp on an air pad, and a hose carries the air to an upright **launch tube**. The foam rockets it comes with slide down **over** that tube. Ours is this one on Amazon, but any stomp launcher with a launch tube works. Just measure **your** tube!

MISSION RULES: YOUR ROCKET MUST...

1. **Slide onto the launch tube easily**, but not wobble around loosely.
2. Have a **closed top**, so the air pushes the rocket up instead of shooting out the top.
3. Have **fins**, so it flies straight.
4. Be **light**: thin walls, no solid chunks. Lighter rockets fly higher!
5. Be named **Rocket – First Name Last Initial** in TinkerCad, so your teacher can find it and print it.

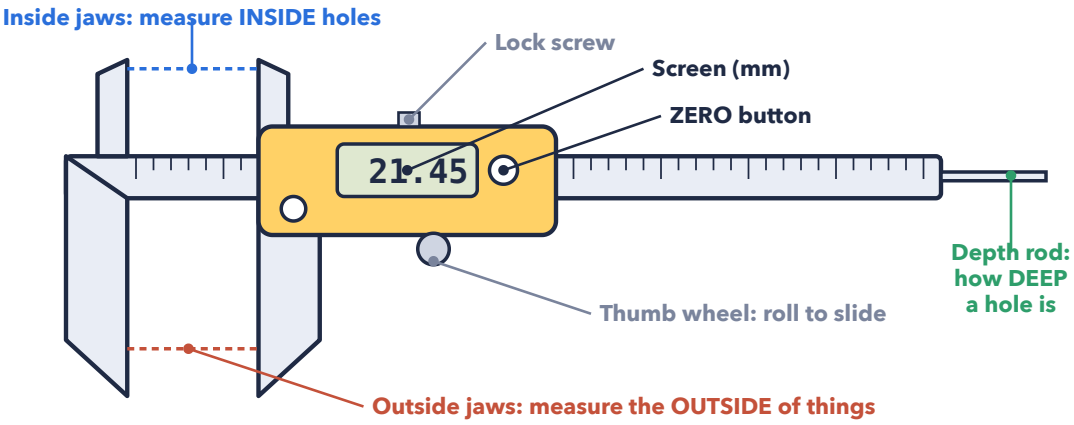
Words to know

Calipers
A tool that measures small things very exactly, down to a hundredth of a millimeter.

Diameter
How wide a circle is, measured straight across through the middle.

Gap (clearance)
A little extra space, so one part can slide over another without getting stuck.

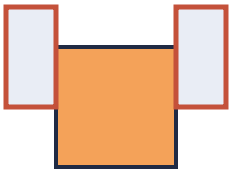
Meet the calipers



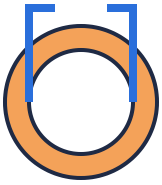
How to use them

- ☐ Press the **on** button. Make sure the screen says **mm** (not inches). If it doesn't, press the **mm/in** button.
- ☐ Close the jaws all the way, then press **ZERO**. The screen should say `0.00`.
- ☐ Open the jaws, put the object in, and close them **gently**. Don't squeeze! Squeezing gives the wrong number.
- ☐ Read the screen. Write down **all** the numbers, like `21.45` mm.

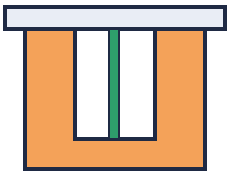
Three ways to measure



Outside
Squeeze the big bottom jaws gently around it.



Inside
Put the small top jaws in the hole, then open them until they touch.



Depth
Rest the end on the edge and slide the rod down to the bottom.

PRACTICE FIRST
Measure these, then check with a neighbor. Did you get the same answer?

OBJECT	WHAT TO MEASURE	MY NUMBER	NEIGHBOR'S NUMBER
Pencil	Outside diameter		
A coin	Outside diameter		
A cup or bottle cap	Inside diameter		

🔍 Measure the launcher

Now measure the real thing. Engineers **measure three times** in different spots, because tubes aren't always perfectly round. Then they find the **average**.

The launch tube

WHAT TO MEASURE	TRY 1	TRY 2	TRY 3	AVERAGE
(A) **Outside diameter** of the launch tube				mm
How far a rocket slides down onto the tube				mm

To find the average: add the three numbers, then divide by 3. Example: (21.4 + 21.6 + 21.5) ÷ 3 = 21.5

A foam rocket (it already works, so let's learn from it!)

WHAT TO MEASURE	MY NUMBER
Inside diameter of its hole (use the inside jaws)	mm
Depth of its hole (use the depth rod)	mm
Total length of the whole rocket	mm
How many **fins** does it have?	

🤔 Think like an engineer

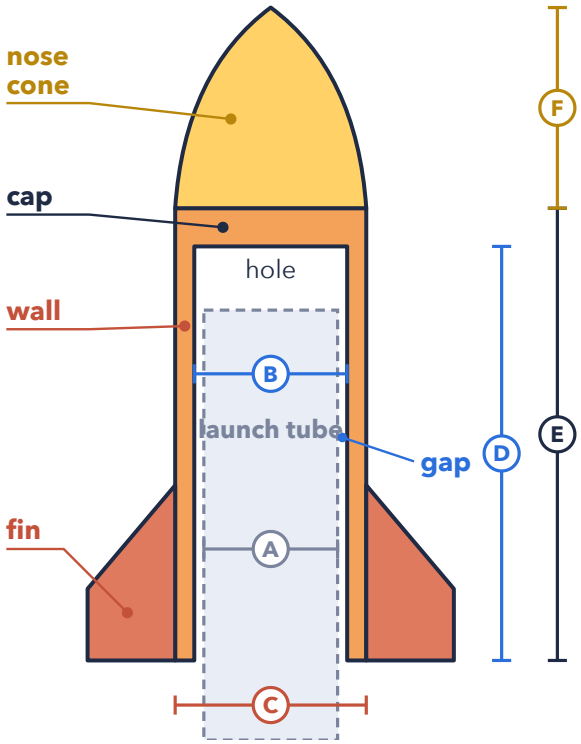
Is the foam rocket's hole bigger or smaller than the launch tube? By how much?

_____ mm. That difference is its **gap**.

Why do you think the hole isn't exactly the same size as the tube?

✏️ Plan your rocket

Here's your rocket cut in half, sitting on the launch tube. Each letter is a size you need to figure out.



(A) Launch tube
`from page 3 (average)` _____

(B) Hole diameter
`A + gap (try 0.8)` _____

(C) Body diameter
`B + 2 walls (try 1.5 each)` _____

(D) Hole depth
`look at the foam rocket!` _____

(E) Body height
`D + cap (at least 2)` _____

(F) Nose cone height
`your choice!` _____

WHY ADD A GAP?

If the hole is **exactly** the size of the tube, the rocket gets stuck. 3D printed holes also come out a tiny bit smaller than designed. A gap of about `0.8` mm is a good start.

WHY ADD 2 WALLS?

The wall goes all the way around, so it's on **both** sides of the hole when you measure straight across.

CHECK YOUR PLAN

B is bigger than A ☐ C is about 3 mm bigger than B ☐ E is bigger than D ☐ E + F is under 150 mm ☐

1 Get set up

- [ ] Open TinkerCad and click **Create** → **3D Design**.
- [ ] **Name it right away!** Click the silly random name at the top left, type **Rocket – First Name Last Initial**, and press `Enter`.

TYPING EXACT SIZES
Click a shape, hover over a white square, click the number that pops up, and type your size from page 4 (decimals like `23.8` are fine!). Press `Enter`.

2 The body

- [ ] Drag a **Cylinder** onto the workplane.
- [ ] Make it (C) wide, (C) long, and (E) tall.

3 The hole

- [ ] Drag another **Cylinder** next to the body. Click **Hole** ◌.
- [ ] Make it (B) wide, (B) long, and (D) tall. Leave it sitting on the workplane, so the hole opens at the bottom.
- [ ] Select the hole **and** the body (hold `Shift` and click both). Click **Align** 品 and click the **two middle** dots that line things up side to side. Don't click the up-and-down row!

4 The nose cone

- [ ] Drag a **Paraboloid** (rounded) or a **Cone** (pointy) onto the workplane. Which one do you think will fly better?
- [ ] Make it (C) wide, (C) long, and (F) tall.
- [ ] Grab the black cone on top and lift it to (E), so it sits right on top of the body.
- [ ] Select the nose and the body, **Align** the two middle dots again.

CHECK
Click the view cube's **Home** button. You should see one smooth rocket shape, with the nose sitting right on top of the body (not floating, not sunk in).

5 The fins

Here's a trick: one long, thin box sticking out on **both** sides of the rocket makes **two** fins at once!

- [ ] Drag a **Box** onto the workplane. Make it **C + 30** long, `2` wide (that's the fin thickness), and `30` tall.
- [ ] Select the box and the body, **Align** the two middle dots. Now it pokes out 15 mm on each side.
- [ ] With just the box selected, press `Ctrl` + `D` (Duplicate). Grab the **curved arrow on the flat part, near the bottom**, and turn the copy **90°**. Now you have 4 fins!

ENGINEER'S CHOICE

Change the size of your fins, or try the **Wedge** shape for slanted fins. Bigger fins fly straighter, but they're heavier. What will you choose? Keep fins at least `1.5` mm thick so they don't snap.

6 Group it

- [ ] Press `Ctrl` + `A` to select everything, then click **Group** (`Ctrl` + `G`).
- [ ] Turn the view cube to look at the rocket **from underneath**. You should see the round hole. The hole also cleans the fins out of the inside!

7 Save it and hand it in 💾

- [ ] Check the name at the top left says **Rocket – First Name Last Initial**.
- [ ] TinkerCad saves by itself. To be sure, click the **TinkerCad logo** (top left) and find your rocket in your designs.
- [ ] Write your sizes from page 4 somewhere safe. If your rocket doesn't fit, you'll need them to fix it!
- [ ] Tell your teacher your rocket is ready to print. 🖨️

🚀 Launch day!

SAFETY RULES

- Never aim the launcher at a person, an animal, or a window. Always point it up!
- Only stomp when your teacher says "Go!" and everyone is standing back.
- Wait for a rocket to land before you pick it up.

The fit test

Slide your rocket onto the launch tube. How does it feel? Circle one:

Too tight 😬 Just right 😃 Too loose 🙃

Too tight? Make your gap bigger. Too loose? Make it smaller. Change (B) and (C), then print again.

Launch log

LAUNCH	HOW DID IT FLY? (STRAIGHT, WOBBLY, SPINNING…)	HOW HIGH OR FAR?
1		
2		
3		

🤔 Think about it

1. Did your rocket fit on the first try? What would you change about your gap?

2. Look at the rocket that flew the highest in class. What was different about it?

3. If you made Rocket 2.0, what would you change, and why?